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Nurture and Nonshared Environment in Cognitive Development

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Behavioral genetic research has demonstrated that shared genetics, not shared environment, makes adults who grew up in the same family similar in personality and psychopathology. The same research affirms the importance of the environment but shows that salient environmental influences in adulthood are not shared by family members; they are unique to the individual. Cognitive traits such as cognitive abilities and educational achievement are thought to be an exception, with half of the environmental variance attributed to shared environmental influences. However, most of this cognitive research has involved children. Here, we show that shared environmental influence on cognitive abilities and educational achievement declines from accounting for 20%–30% of the variance in childhood to 10%–20% in adolescence and to near 0% by early adulthood. Educational attainment (years of schooling) shows lasting shared environmental influence (30%) carried over from decisions made in adolescence to go to university, which shows the greatest shared environmental influence (47%). We conclude that specific cognitive abilities as well as general cognitive ability show moderate shared environmental influence in childhood when children live at home, but this influence disappears as young people make their own way in the world. We propose that random endogenous processes are responsible for nonshared environmental influences on adult cognitive abilities. We discuss the far-reaching implications for understanding the environmental causes of individual differences in cognitive abilities in adulthood.

Keywords: nonshared environment, cognitive abilities, educational achievement, randomness, genetic nurture

Supplemental materials: <https://doi.org/10.1037/rev0000551.supp>

Twin and adoption studies have shown that inherited deoxyribonucleic acid differences account for about half of the variance in behavioral traits (Polderman et al., 2015). The other half of the variance led to a finding with at least as far-reaching implications. Because family members share both genes and environment, the rationale for twin and adoption research was to disentangle the effects of nature and nurture in family resemblance. Environmentalism, which dominated the behavioral sciences for most of the twentieth century (Pinker, 2003), assumed that behavioral traits run in families for environmental reasons, ignoring the fact that first-degree relatives are 50% similar genetically. On the contrary, genetic research has shown that familial resemblance for personality and psychopathology in adulthood can be explained entirely by the genetic similarity between relatives, although some shared environmental influence is found in childhood and adolescence (Burt, 2009, 2024). The environment is important, accounting for the other half of the variance,

but it is not nurtured in the sense of a shared family environment. The salient environmental influences are not shared by children growing up in the same family. The use of twin and adoption methods to estimate shared environmental influence is discussed in Supplemental Material 1.

The importance of environmental differences within rather than between families was first highlighted in 1976 by Loehlin and Nichols (1976) who proposed that environmental effects seem to “operate almost randomly” (p. 92), although earlier passing references can be found (Woodworth, 1941). In 1981, this phenomenon was labeled *nonshared environment* (Rowe & Plomin, 1981). In 1987, a review concluded that nonshared environment is the major source of environmental variance for personality and psychopathology (Plomin & Daniels, 1987), which has become one of the most replicated findings in behavioral genetics (Plomin et al., 2016).

In contrast, general cognitive ability yielded substantial estimates of shared environmental influence, accounting for as much as 30% of the variance. However, the 1987 review noted: “Although it has been thought that cognitive abilities represent an exception to this rule, recent data suggest that environmental variance that affects IQ is also of the nonshared variety *after adolescence*” (Plomin & Daniels, 1987, p. 1). This review was published with 32 commentaries, which did not disagree with the revolutionary finding that environmental influence was largely nonshared for personality and psychopathology. However, the suggestion that this was also true for cognitive abilities after adolescence met with incredulity (Jensen, 1987).

It is important to know whether environmental influences on cognitive traits are nonshared after adolescence because research trying to identify environmental influences on cognitive traits has assumed that the salient environmental influences are shared

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Robert Plomin played a lead role in conceptualization, supervision, and writing—original draft. Kaito Kawakami played a lead role in software and visualization and a supporting role in writing—original draft.

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by siblings growing up in the same family, such as influences of parents, families, and schools. Associations between such family-wide environmental measures and cognitive traits have been found in childhood independent of genetics (Mackintosh, 2011). However, if environmental variance after adolescence is nonshared, little is known about the environmental factors responsible for individual differences in cognitive traits in adulthood when they matter for the success of society.

The present review has three goals. First, we review the extensive research relevant to the hypothesis that shared environmental variance for general cognitive ability diminishes in adolescence and is near zero by young adulthood. However, other cognitive traits—a broad term which we use to encompass specific cognitive abilities, educational achievement, and educational attainment—might have different developmental trajectories, so the second goal is to review shared and nonshared environmental estimates for cognitive traits other than general cognitive ability. Third, we consider the implications of these findings for understanding the environmental causes of cognitive traits in adulthood.

Throughout this review, to focus on the shared and nonshared environments, we reluctantly leave aside the major source of individual differences in cognitive traits: genetics, which accounts for about half of the variance of cognitive traits. Specific genetic issues, such as the distinction between additive and nonadditive genetic variance, assortative mating, and genotype-environment correlation and interaction are discussed in Supplemental Material 1, as they also concern the estimation of shared and nonshared environment (Knopik et al., 2017). We must also neglect the deoxyribonucleic acid revolution, which has transformed research in behavioral genetics (Plomin, 2023), although in the section on educational attainment, we mention a topic of recent interest called “genetic nurture,” thought to be an environmentally mediated genetic effect (Kong et al., 2018). This long list of important omissions is a sign of the vitality of the field, but yielding to the temptation to discuss these topics would dilute our focus on nurture and nonshared environment. While we acknowledge the importance of these issues, the purpose of the present review is to focus on nurture and nonshared environment. For in-depth discussions of these issues, we direct readers elsewhere (Nivard et al., 2024).

Two other general caveats are in order. First, the distinction between *what is* and *what could be* warrants emphasis. Genetic research describes genetic and environmental sources of variance as they exist in a particular population at a particular time. The research does not predict what could be in other populations at other times. For example, parents *can* without doubt affect their children’s cognitive development, especially adversely, but the question addressed by genetic research is the extent to which individual differences in parenting *do* have a shared environmental effect on individual differences in their children’s cognitive traits. For the same reason, finding that environmental variance is nonshared does not imply that family or school interventions can have no effect. A second caveat is that the research we review cannot be safely generalized beyond the samples studied, which means that these studies are limited to the normal range of families who participate in research, which does not include, for example, the extremes of neglect or abuse. We limit our review to the normal distribution of cognitive traits because much less research is available at the extremes. However, common disorders are likely to be the

quantitative extremes of the same genetic and environmental factors responsible for normal distributions (Plomin et al., 2009).

Transparency and Openness

Data were visualized in R version (Version 4.2.3), using the ggplot2 package (R Core Team, 2013; Wickham et al., 2019). This review project was not preregistered. Data and code used to produce visualizations are available on https://github.com/kaito-kawakami/nse_cog.

General Cognitive Ability

Twin and adoption studies consistently converge on the conclusion that environmental influences on general cognitive ability (*g*, aka intelligence, IQ) are substantially due to nurture as well as nature. However, as indicated in the original 1987 review (Plomin & Daniels, 1987) and substantiated in subsequent research (McGue et al., 1993), a developmental perspective suggested that these shared environmental influence decline in adolescence and disappear in adulthood. In this section, we summarize the results of our review, with more details provided in Supplemental Material 2.

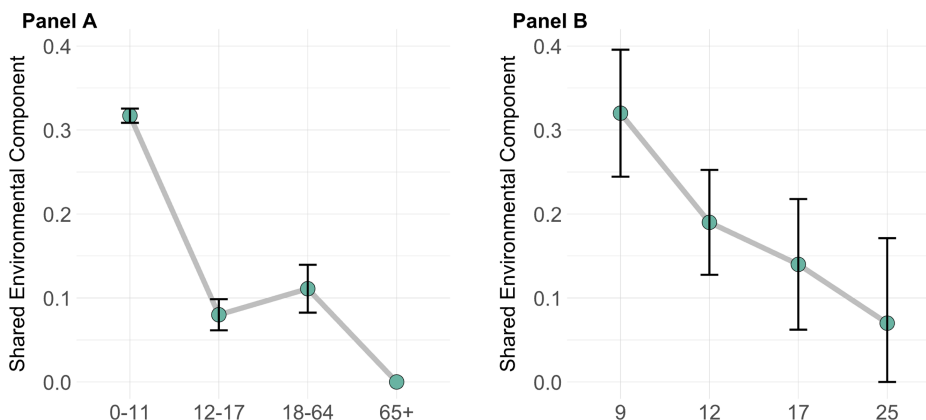
By far, most data come from the twin design, which estimates shared and nonshared environment influence indirectly as residual variance not accounted for by genetics. Familial resemblance not explained by genetics is attributed to shared environment and the rest of the variance is ascribed to nonshared environment and error of measurement (see Supplemental Material 2 for more details).

Figure 1 summarizes twin study estimates of shared environmental influence by age, with more details in Supplemental Material 2. Panel A of Figure 1 shows results extracted from a 2015 meta-analysis of twin correlations across hundreds of “higher-level cognitive functions” for more than 200,000 twin pairs (Polderman et al., 2015). Estimates of the shared environment were 32% for children (below age 12), 8% for adolescents (12–17 years), and 0% in adulthood (18–64 and 65+). As shown in Panel B, a mega-analysis of 11,000 twin pairs in six recent twin studies with standard measures of *g* demonstrated a significant decrease in shared environmental influence from 33% in childhood (average age 9 years) to adolescence (18% at 12 years; 16% at 17 years). A recent extension of the largest of these twin studies to age 25 showed a further decline to 8% (Malanchini et al., 2021). For both Panels A and B, the decline in shared environmental influence is matched by an increase in heritability rather than an increase in nonshared environmental influence. The developmental increase in heritability for general cognitive ability is one of the most replicated findings in behavioral genetics, which reflects increasing identical twin correlations and decreasing fraternal twin correlations (Haworth et al., 2010).

The most direct test of the hypothesis that family resemblance is caused by shared environment is the correlation between adoptive relatives, genetically unrelated family members who live together, most notably adoptive parents and their adopted children and adoptive siblings. Adoption studies before the 1980s yielded an average *g* correlation of 0.24 between adoptive parents and their adopted children, suggesting that parents have substantial shared environmental effects on their children (Bouchard & McGue, 1981). However, three more recent adoption studies found correlations of about 0.10 in childhood and near zero in adolescence and adulthood (Horn & Loehlin, 2010; Plomin et al., 1997; Willoughby, McGue,

Figure 1

Twin Study Estimates of Shared Environmental Influence for General Cognitive Ability Decline From Childhood to Adulthood



Note. Panel A depicts estimates extracted from twin correlations in a meta-analysis of more than 200,000 twin pairs for “higher-level cognitive functions” (Polderman et al., 2015). Panel B shows estimates from a mega-analysis of g from 11,000 twin pairs (Haworth et al., 2010) with an extension to age 25 (Malanchini et al., 2021). Error bars indicate 95% confidence intervals. See the online article for the color version of this figure.

Iacono, & Lee, 2021). These more recent adoption studies cast doubt on the conclusion that parents’ g -related traits are transmitted environmentally to their children. Importantly, parents’ g does not capture all the shared environmental influences relevant to their children’s cognitive development. The resemblance between adoptive siblings, genetically unrelated children reared in the same adoptive family, directly estimates the importance of all shared environmental influences that affect siblings growing up together in the same family. Studies of adoptive siblings before the 1980s yielded a g correlation of 0.32, indicating that a third of the total variance of g could be attributed to environmental influence shared by siblings (Bouchard & McGue, 1981).

However, no one noticed that this correlation was based on preadolescent children. The first study of postadolescent adoptive siblings (average age 18 years) reported a correlation of -0.03 (Scarr & Weinberg, 1978), a finding confirmed in two additional studies of postadolescent adoptive siblings, which yielded an average g correlation of -0.01 , in contrast to a correlation of 0.25 from 11 recent studies of preadolescent adoptive siblings (McGue et al., 1993). The strongest evidence for the decline of shared environmental influence comes from three subsequent longitudinal studies of adoptive siblings, which confirmed the cross-sectional findings in that average adoptive sibling g correlations are 0.25 in childhood, 0.13 in adolescence, and 0.05 in adulthood (Willoughby, McGue, Iacono, & Lee, 2021; see Supplemental Material 2 for details). These findings are also supported by a meta-analysis of longitudinal data through adolescence from recent adoption and twin studies (Briley & Tucker-Drob, 2013).

Dramatic confirmation of the hypothesis that shared environmental influence on g is near zero in adulthood comes from adult identical twins who had been reared apart: they are as similar as adult identical twins who had been reared together (McGue et al., 1993). In other words, identical twins growing up in the same family are no more similar for g than identical twins growing up in different families.

In summary, although general cognitive ability has been thought to be an exception to the rule that environmental influence is nonshared, it is now clear that shared environmental influence on g has little long-term impact after children leave home and make their own way in the world. In other words, environmental influences that account for almost half the variance of general cognitive ability are, in the long run, not caused by nurture in the sense of systematic environmental effects of parenting or schools shared by siblings growing up together. The salient environmental influences responsible for individual differences in general cognitive ability in adulthood are nonshared, making siblings who have grown up together no more similar than if they had been reared in completely different families.

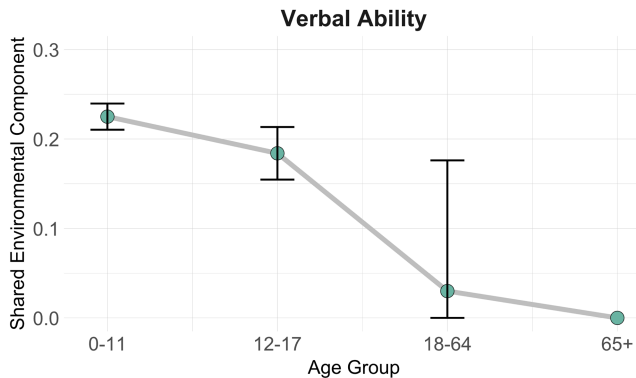
Specific Cognitive Abilities

Does this developmental pattern of results for g extend to specific cognitive abilities including verbal traits like vocabulary and verbal fluency and nonverbal traits like memory and spatial ability? Previous reviews of the genetics of specific cognitive abilities were limited to twin research and are more than 30 years old (Plomin, 1988). Most of the twin studies available at that time involved samples too small to provide reliable estimates of shared environmental influence and they did not consider age. In this section, we review newer twin and adoption studies from a developmental perspective, with more details provided in Supplemental Material 3.

Estimates of shared environmental influence that we extracted from two recent reviews of twin studies indicate that, like g , specific cognitive abilities show substantial shared environmental influence in childhood that disappears in adulthood. The first review was a 2015 meta-analysis that estimated shared environmental influence for broad categories of verbal, memory, and numerical ability (Polderman et al., 2015). As shown in Figure 2, estimates were about 20% in childhood and adolescence for verbal ability but declined to zero in adulthood. Not included in Figure 2 are results for memory ability because it was not assessed in adolescence, and

Figure 2

Twin Study Estimates of Shared Environmental Influence on Verbal Ability



Note. Twin study estimates of shared environmental influence on broad categories of verbal ability were extracted from a 2015 meta-analysis (Polderman et al., 2015). Error bars depict 95% confidence intervals. The lower bound of the 18–64 age group was negative and set to zero in this illustration. See the online article for the color version of this figure.

numeric memory because it was only assessed in childhood. Shared environment estimates for memory ability were 0.05 at ages 0–11 and zero both at ages 18–64 and 65+ (Table S1, Supplemental Material 3).

Twin correlations obtained from a more refined 2022 meta-analysis of adequately powered twin studies of 11 specific cognitive abilities (Procopio et al., 2022) also found a general downward developmental trend, although for six abilities data were available at only one or two ages (Figure 3 and Table S2, Supplemental

Material 3). We note that while the focus of the Procopio et al. (2022) review was on the heritability of specific cognitive abilities, we were able to calculate estimates of shared environment using twin correlations obtained through personal correspondence (Table S2, Supplemental Material 3). As shown in the top left-hand corner of Figure 3, the average shared environment estimates across the 11 abilities were 40% in early childhood, 13% in middle childhood, 12% in adolescence, 3% in adulthood, and 1% in older adulthood.

Much less research is available for specific cognitive abilities using the adoption design. As described in Supplemental Material 3, the average adoptive sibling correlation in adolescence across specific cognitive ability measures in five studies was 0.10, but the results do not allow comparisons across age (Horn & Loehlin, 2010; Willoughby, McGue, Iacono, & Lee, 2021). In four studies of adoptive parents and their adopted children, the average correlation was 0.05 across childhood and adolescence, indicating little shared environmental transmission from parents to offspring for specific cognitive abilities (Horn & Loehlin, 2010; Plomin et al., 1997; Scarr & Weinberg, 1978; Willoughby, McGue, Iacono, & Lee, 2021).

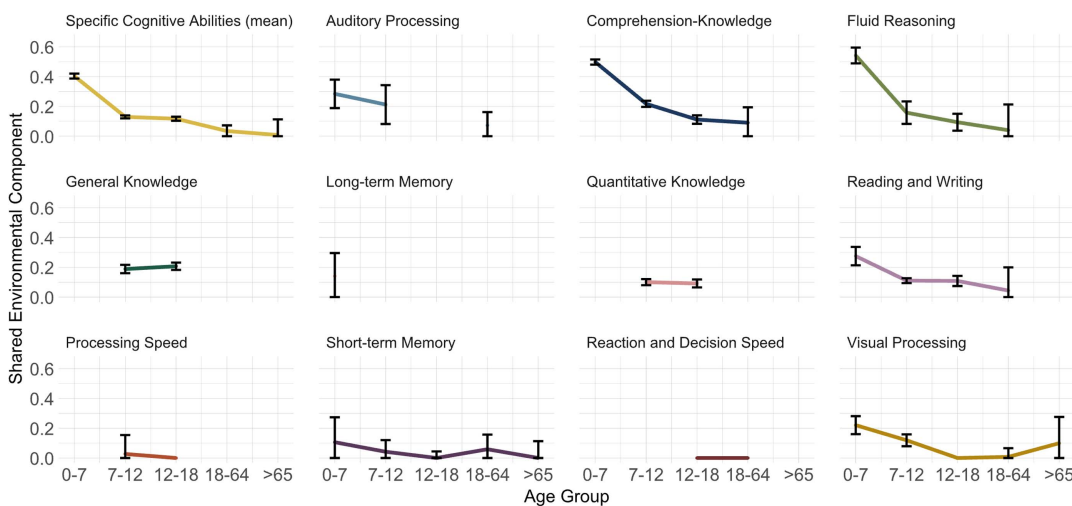
In summary, although more adoption research is needed, twin studies of specific cognitive abilities, like *g*, reveal substantial shared environmental influence in childhood that fades by early adulthood.

Educational Achievement

Educational achievement is meant to assess what is taught at school, in contrast to cognitive abilities which connote inherent intellectual capabilities. Although *g* and school achievement are strongly correlated (~ 0.70), noncognitive traits such as conscientiousness contribute independently toward achievement (Malanchini et al., 2020).

Figure 3

Twin Study Estimates of Shared Environmental Influence on Diverse Specific Cognitive Abilities Decline From Childhood to Adulthood



Note. Twin study estimates of shared environmental influence on specific cognitive abilities were calculated using twin correlations obtained through personal correspondence from a 2022 meta-analysis focused on genetic influence (Procopio et al., 2022). Error bars depict 95% confidence intervals (CIs), with negative CI values set to zero for illustrative purposes. See the online article for the color version of this figure.

Using national test results for literacy, numeracy, and science, large twin studies in the United States at age 13 (Loehlin & Nichols, 1976) and in the United Kingdom at age 16 (Shakeshaft et al., 2013) found shared environmental estimates of about 30%. However, a large Dutch study yielded an estimate of only 8% for a total score on a national test administered at age 12 (de Zeeuw et al., 2016), although an earlier Dutch analysis using the same test at the same age yielded an estimate of 27% (Bartels et al., 2002; Supplemental Material 4 includes more detail about studies in this section).

A meta-analysis of 28 reports with diverse measures from 11 twin studies found an average shared environmental estimate of about 12% (de Zeeuw et al., 2015). The review was limited to primary school and most of the samples were in the first 3 years of school, but this does not seem to be the sole reason for the lower estimate because the only longitudinal twin analysis found estimates of about 20% that did not vary systematically across the school years (Rimfeld et al., 2018).

In summary, estimates of shared environment in twin studies vary from about 10% to 20%. The most compelling evidence for the importance of a shared environment for educational achievement is a correlation of 0.24 for 534 pairs of adoptive siblings for scores on a national examination at the end of primary school in Denmark (Ludeke et al., 2021).

Figure 4 shows that twin study estimates of shared environmental influence did not differ much across subjects. Nor did the results vary systematically from the early school years to secondary school (Supplemental Material 4). Although these estimates vary widely across studies, the average shared environment estimate from Figure 4 is 21%.

Does shared environmental influence on educational achievement fade after secondary school, as it does for cognitive abilities? Two U.K. reports suggest that it does. In the United Kingdom, compulsory secondary schooling ends at age 16. A gatekeeper for enrolling at university is a 2-year university preparatory course called Advanced Level (A-level). A twin analysis of A-level examination scores yielded shared environment estimates of only 2% for science, technology, engineering, and mathematics subjects and 11% for humanities (Rimfeld et al., 2016). Even more surprisingly, university grades showed no significant shared environmental influence (Smith-Woolley et al., 2018).

Although these results are based on a single U.K. study of A-level scores and university grades, the evidence as it stands suggests that educational achievement, like cognitive abilities, shows substantial shared environmental influence in primary and secondary education (about 20%) that fades during tertiary education.

Educational Attainment

If shared environmental influence on educational achievement disappears in adulthood, one might suppose that educational attainment—years of schooling, which can only be assessed in adulthood after schooling has been completed—would also show no shared environmental influence. But this is not the case. Because educational attainment can be assessed with a single item and is used as a demographic index in research, large samples of twins are available. In a mega-analysis of educational attainment from 28 twin cohorts with 81,894 twin pairs, shared environmental influence was

estimated as 31% (Silventoinen et al., 2020). An earlier meta-analysis yielded an estimate of 36% (Branigan et al., 2013).

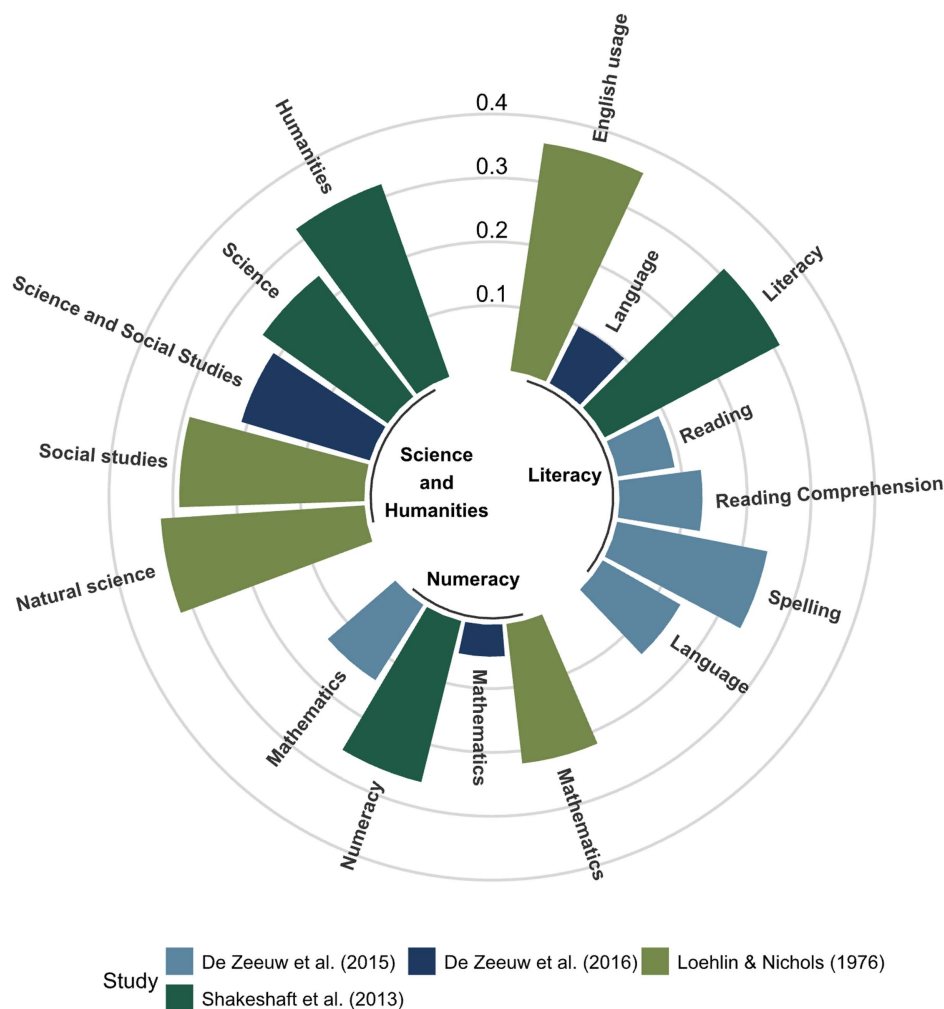
The only adequately powered study of educational attainment using the adoptive sibling design strongly supports these twin results, reporting a correlation of 0.37 for adoptive siblings (McGue et al., 2017). In contrast, across eight adoption studies, the average weighted correlation for educational attainment of adoptive parents and their adopted children is 0.08, indicating that parents' educational attainment has little shared environmental effect on their children (See Supplemental Material 5 for details and for a discussion of the effects of assortative mating, which is substantial for educational attainment).

Because secondary school is mandatory in developed countries, most of the variance in educational attainment comes from whether individuals go to university (Rietveld et al., 2014). Although educational attainment can only be assessed in adulthood, it is the outcome of performance and choices in secondary school. In the United Kingdom, going to university depends on making the decision to pursue university education by enrolling in the requisite A-level course at age 16. A-level enrollment showed the highest shared environmental estimate (47%) of all education-related variables (Rimfeld et al., 2016). The subsequent decision to actually enroll at university also showed substantial shared environmental influence (36%; Smith-Woolley et al., 2018). It seems likely that these gatekeepers for going to university are responsible for the shared environmental influence on educational attainment. Although no comparable data for these gatekeepers of educational attainment are available for countries other than the United Kingdom, the mega-analysis of educational attainment mentioned earlier found similarly substantial shared environmental influence in Europe, North America and Australia, and East Asia (Silventoinen et al., 2020).

Educational attainment has been a focus of a topic of current interest in genomic research called "genetic nurture" (Kong et al., 2018). Genetic nurture refers to the phenomenon in which parents' genotypes predict offspring traits independent of offspring genotypes. This represents the portion of the parental genome that offspring do not inherit, which nonetheless affects the offspring's phenotypes through some unmeasured environmental factors such as the rearing environment. Using the educational attainment polygenic score (Okbay et al., 2022) as the prime example, parents' polygenic scores for educational attainment independent of offspring polygenic scores predict unmeasured characteristics of the parents that relate environmentally to the educational attainment of their offspring (Wang et al., 2021). For educational attainment, the salient parental characteristics appear to be socioeconomic status and cognitive ability (Willoughby, McGue, Iacono, Rustichini, & Lee, 2021). Because these characteristics are the same for all children in the family, they must relate to offspring educational attainment via shared environment. We suggest that genetic nurture has been primarily found for educational attainment and cognitive ability because these are the behavioral domains that show the most shared environmental influence. However, genetic nurture is a weak index of shared environment because it only captures shared environmental influence that is associated with parental polygenic scores. In this sense, genetic nurture is a genetically indexed environmental effect rather than an environmentally mediated genetic effect, as it is often described. In other words, genetic nurture is less about

Figure 4

Circular Bar Chart of Shared Environmental Estimates for Educational Achievement Across Subjects



Note. Circular bar chart of shared environmental estimates for educational achievement across subjects (grouped by literacy, numeracy and science) assessed using nationwide tests in three large twin studies (de Zeeuw et al., 2016; Loehlin & Nichols, 1976; Rimfeld et al., 2018) and in a meta-analysis of multiple reports from 11 studies (de Zeeuw et al., 2015). See the online article for the color version of this figure.

nurture in terms of shared environment than it is about the “nature of nurture,” genetic influence on the environment (Plomin & Bergeman, 1991).

We conclude that, although environmental influences shared by parents and their offspring have only a modest effect on educational attainment, environmental influences shared by siblings, which include parental effects independent of their educational attainment, have a substantial impact (about 30%). Although this conclusion about educational attainment might seem at odds with the conclusion that shared environmental influences for cognitive traits disappear in adulthood, educational attainment is the result of shared environmental influences on diverging pathways in adolescence that lead to tertiary education. In other words, shared environmental influence on adult educational attainment reflects the lasting effects

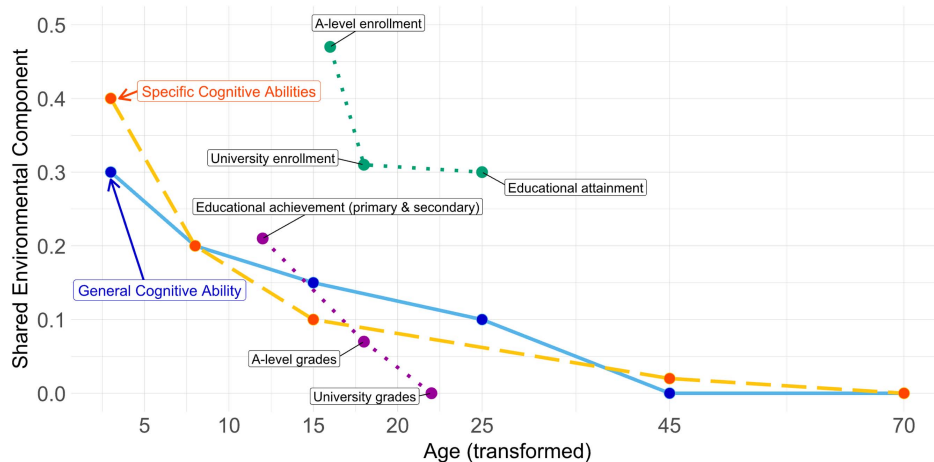
of shared environment in adolescence rather than experiences in adulthood.

Conclusions

Three conclusions emerge from these findings, as illustrated in Figure 5. First, shared environmental influence for general and specific cognitive abilities declines from about 30% in childhood to 10% in adolescence and to near 0% by early adulthood. Second, for educational achievement, estimates of shared environment are about 20% throughout primary and secondary school but disappear in tertiary education. Third, what stands out is the substantial estimate of shared environment for the decision to take the 2-year A-level preparatory course for university (47%), which drives shared

Figure 5

Summary of Shared Environmental Estimates for Cognitive Traits in Childhood, Adolescence, and Early Adulthood



Note. For interpretability, ages 30–70 have been scaled to fit into a compressed range, equivalent in width to that of ages 0–25. See the online article for the color version of this figure.

environmental influence on the decision to enroll at university (31%) and, ultimately, on educational attainment (30%).

In other words, for cognitive traits, moderate shared environmental influence is found in childhood when children live at home under the influence of their parents and attend the same schools, but the influence of shared environment disappears by late adolescence as young people make their own way in the world. The only exception is educational attainment, which is nonetheless driven by decisions made in adolescence to go to university. In summary, environmental influences that affect cognitive traits in adulthood are almost entirely nonshared. In adulthood, the phrase *cognitive traits* is limited to cognitive abilities rather than educational achievement because formal education typically ends in early adulthood.

Implications

These findings have far-reaching implications for science and society. Understanding environmental influences on cognitive abilities, society's true intellectual capital, is important because their societal impact emerges as young people enter the workforce. Yet hardly anything is known about these nonshared experiences. One reason for this lack of knowledge is that nearly all research on the environmental causes of cognitive differences has assumed these influences are shared (Scarr, 1997), an assumption that our review indicates is wrong.

What are the nonshared experiences responsible for differences in cognitive abilities among adults? Any experience in adulthood can be considered a source of nonshared environmental influence, including accidents, illnesses, further education, self-improvement, jobs, parents, spouses, friends, peers, social media, sports, and religion. Two general methods are available for identifying nonshared environmental effects controlling for genetic and shared environmental effects. The touchstone for identifying nonshared environmental influences is differences within pairs of identical twins. Because identical twin pairs inherit identical deoxyribonucleic acid,

their cognitive differences can only be due to nongenetic factors that make siblings different, thus bypassing genetic confounding. Another method is multivariate genetic analysis which can parse correlations between environmental measures and cognitive abilities into genetic, shared environmental, and nonshared environmental components. Neither of these approaches has been used to identify nonshared environmental influences on cognitive abilities in adulthood.

Although there are many candidates for nonshared environmental influence on cognitive abilities in adulthood and methods exist to confirm their candidacy, two sets of findings bode ill for this enterprise. The first omen is that 30 years of research trying to identify the causes of nonshared environment for psychopathology and personality in a line-up of the usual suspects, especially parenting, has not identified the culprits (Plomin, 2024). This research on the usual suspects is analogous to the candidate gene phase of genomics in the 1990s, which investigated a handful of genes thought to be associated with a trait. With the advent of genome-wide association analysis, the candidate gene approach was superseded by genome-wide association, a polygenic model that recognized that the ubiquitous heritability of complex traits is caused by many genetic variants with very small effect sizes (Visscher et al., 2021). Similarly, nonshared environment might be due to many events each with very small effect size, a "polyenvironmental" model, which is becoming feasible with technological advances that can capture microenvironmental events in real time, such as wearable devices and online footprints (von Stumm & d'Apice, 2022). It is an empirical issue, not yet investigated, whether these new technologies can capture experiences that systematically predict nonshared environmental effects on cognitive abilities in adulthood after controlling for gene–environment correlation.

The second omen is that nonshared environmental influences on cognitive traits appear to be unstable. From childhood through adolescence, longitudinal twin studies indicate that age-to-age stability of educational achievement and general cognitive ability is primarily due to genetics and secondarily to shared environmental

influences (Briley & Tucker-Drob, 2013; Rimfeld et al., 2018). Nonshared environment contributes to age-to-age change, not continuity. In other words, nonshared experiences that impact cognitive abilities in adulthood are not the residue of nonshared experiences earlier in life. Although longitudinal genetic research on cognitive abilities in adulthood is scarce, extant research suggests that genetics accounts for stability and nonshared environment contributes to age-to-age change in adulthood (Rimfeld et al., 2018).

Are these unstable nonshared environmental effects merely error of measurement? Although there are many approaches to measurement error, the most conservative is short-term test–retest reliability, which indexes error as change (Guttman, 1945). Cognitive abilities, especially intelligence tests, are highly reliable, with 2-week test–retest reliabilities exceeding 0.90 (Schuerger & Witt, 1989), which means that measurement error so defined accounts for less than 10% of the variance. If the variance of cognitive abilities is attributed 50% to genetics, 0% to shared environment, and 10% to measurement error, then nonshared environment accounts for the remaining 40% of the variance beyond short-term measurement error.

A reconceptualization of nonshared environmental factors that can address this instability begins with the proposition that they are random (Plomin, 2024), which has been usefully defined as unpredictability (Eagle, 2005). Accidents and illnesses are classic examples of random sources of nonshared environmental influence that can make identical twins different. Microenvironments, as mentioned earlier, are also likely to appear random in the sense of being unpredictable.

The second component of this reconceptualization adds randomness to the consideration of endogenous processes. The search for nonshared environmental influences has focused on exogenous events external to the individual such as parenting, accidents, and microenvironments. Instead, random endogenous noise generated by complex biological systems independent of exogenous events has been proposed as the major source of nonshared environmental influence, summarized as “the intrinsic stochasticity of molecular processes” (Tikhodeyev & Shcherbakova, 2019, p. 259). The best example of a random endogenous nonshared environmental process that differs within pairs of identical twins is somatic mutations, noninherited mutations that can occur with each cell division (Jonsson et al., 2021). However, noise can come from any endogenous process, such as stochastic developmental changes in neuroanatomical structures (Mitchell, 2018) and epigenetics (Li et al., 2022), as well as psychological processes (Plomin, 2024). In relation to psychological processes, the neuroscience of consciousness (Seth, 2021) and mindfulness (Williams & Kabat-Zinn, 2013) could be helpful in understanding these random transient experiences. For example, randomness may be a basic feature of perception, such that all input is filtered through a kaleidoscope of cognitive noise. Under this model, even identical twins experiencing identical environments could be affected in different ways. Proving that these are random events with causal effects will be difficult, but it is an important direction for research. Randomness has increasingly been documented in fundamental processes throughout science (Carroll, 2020) from physics (Mack, 2020) to economics (Kay & King, 2020).

A general scientific implication of this radical reconceptualization of nonshared environment is the need to transition from scientists’ strongly held belief in determinism (Sapolsky, 2023) to a probabilistic perspective that embraces randomness (Smith, 2011). Although

randomness limits our ability to predict and control the nonshared environmental causes of individual differences in cognitive abilities in adulthood, we can at least understand the processes, such as biological and psychological noise, by which the nonshared environment has its effects. Because multiple random processes are unlikely to have the same pattern of effects across individuals, this understanding might require pivoting from nomothetic strategies that look for average effects in the population to idiographic approaches that investigate idiosyncratic experiences (Franck, 1986).

A profound societal implication of this reconceptualization impacts parenting. Parents have much less control of their children’s outcomes than the thousands of parenting books claim, for two reasons (Plomin, 2019). First, genetics is the major systematic force making children who they become as adults, but parents do not control their genetic endowment to their children. The second reason is specific to this reconceptualization of nonshared environment: parents do not have the levers to control how their children turn out as adults if environmental effects are random. While an absence of parenting altogether would manifest as neglect, differences in the normal variation of parenting do not appear to contribute toward systematic differences in behavioral outcomes in their children. The cognitive domain offers some consolation in the finding that shared environmental factors, presumably parents and teachers, make a difference in their children’s cognitive abilities and educational achievement, although these effects wane in adolescence and disappear in adulthood. One outcome for which shared environment has an effect lasting into adulthood is the decision to go to university, a decision that casts a long shadow beyond educational attainment, employment, and income to encompass expanded social networks and personal growth (Collini, 2012).

This reconceptualization also has implications for understanding our adult selves. It suggests that we are not the same person from day to day as our cognitive abilities are buffeted by random experiences beyond our control. It is difficult to accept this view because our personal experience of our cognitive strengths and weaknesses is overwhelmingly an experience of permanence. However, this sense of stability occurs because genetics is the systematic, stable force responsible for individual differences in cognitive abilities in adulthood after the impact of earlier shared environment has disappeared.

If our hypothesis is correct that these experiences involve random endogenous processes, it will be difficult to identify the salient nonshared environmental influences on adult cognitive abilities. Nonetheless, because nonshared experiences account for nearly as much of the variance as genetics, it is important to shed light on this dark matter of environmental influences on cognitive abilities in adulthood.

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